

Assignment 2 – Document databases

Description

We will use MongoDB Community Server 8.0.17 and the following collections and documents:

Collection: Businesses

Documents:

```
{_id, bname, is_open, bike_parking, good_for_dancing, restaurants_good_for_groups,
restaurants_price_range, restaurants_take_out, restaurants_delivery,
restaurants_counter_service, restaurants_table_service, restaurants_reservations, drive_thru,
wheelchair_accessible, by_appointment_only, dogs_allowed, happy_hour, byob, corkage,
outdoor_seating, accepts_insurance, caters, coat_check, good_for_kids, open_24_hours, has_tv,
business_accepts_credit_cards, business_accepts_bitcoin, categories: [...], location: {address,
state, city, zipcode, latitude, longitude}, hours: {0 : {opening, closing}, 1 : { opening, closing},
...}}
```

Where categories is an array with the names (not the ids) of the categories, location contains all the location information, and hours contain, for each day of the week, the opening and closing hours,

Collection: Users

Documents:

```
{_id, uname, yelping_since, useful, funny, cool, fans, compliment_hot, compliment_more,
compliment_profile, compliment_cute, compliment_list, compliment_note, compliment_plain,
compliment_cool, compliment_funny, compliment_writer, compliment_photos, elite_years: [...],
friends: [...]}
```

Where elite_years is an array with the years the user was elite, and friends is an array with the ids of the users that are friends of the user at hand.

Collection: Reviews

Documents:

```
{_id, user_id, business_id, stars, review_date, useful, funny, cool}
```

If a piece of data is not present, the field should be suppressed from the document. For instance, the document of a user that has does not have any fans should not contain a fans field.

The MySQL database will be the same one we used in Assignment 1. For the aggregation queries, use the JSON data available in myCourses.

Your tasks

- 1.- Provide a program that loads Businesses from MySQL to MongoDB. Check the grading software and software templates for more info.
- 2.- Provide a program that loads Users (but not friends) from MySQL to MongoDB. Check the grading software and software templates for more info.
- 3.- Provide a single aggregation query that retrieves the following data. You must create the following indexes:

```
db.Reviews.createIndex({ business_id: 1 })
```

db.Reviews.createIndex({ user_id: 1 })

3.1.- Open businesses that have received more than two hundred and fifty reviews, that belong to more than five categories, and that are closed on Saturdays and Sundays.

3.2.- Top-15 most prolific users who are friends of more than two-hundred and fifty other users and have not reviewed any Mexican businesses. Reviews must be grouped by category.

3.3.- Reviews written by users who started yelping in June 2015 such that there are posterior reviews by the same user, with the exact same stars as well as funny, cool and useful votes. The posterior reviews were written on January 1st, 2020. Retrieve (original id, posterior id) pairs.

3.4.- Top-5 users such that they have reviewed at least five Bookstores with four or more stars.

3.5.- Users whose name contains either "ch" or "th" (case insensitive), who have reviewed more than twenty businesses that belong to categories Korean or Seafood, who have no friends, and who have not reviewed any Mexican businesses after 2020. The Korean or Seafood businesses must be counted.

3.6.- Users who started yelping between January 2021 and May 2021, who have reviewed more than three businesses in Arizona, who are friends of at least two other users, and who have never been elite users.

3.7.- Users with at least seventy-five friends, and who have reviewed at least one business such that the review got more than three funny and cool votes. The users started yelping strictly after all their friends started yelping.

3.8.- Users with more than 50 fans and more than five friends, and all their friends have never written more than three five-star reviews. The users have never reviewed a business located in Largo.

3.9.- Businesses reviewed by all the reviewers of Garage Burger Bar & Grill and the average number of stars received.

3.10.- Korean businesses that, during weekdays, are open at the exact same times or longer as Bayou Daiquiris. For the Korean businesses, provide the total number of useful votes.

There are memory and time requirements. Check the grading software and query templates for more info.

Submission instructions

- Use the software template(s) and grading software provided in myCourses.
- Submit a single ZIP file to myCourses that must be named as your RIT user, e.g., crrvcs.zip. Do not include '@rit.edu.' The file must contain folders named as the folders provided in the software template(s) containing your queries, Gradle or Poetry project(s).
- Everything will be graded on a Linux machine, so you must always use the exact names provided in this document, software template(s) and grading software.

Grading rubric

- All test cases in the grading software must pass with some restrictions like main memory and total running time.